Real-Time Stream Processing Assignment

Course:	ENGR 5785G: Real-time Data Analytics for IoT
Topic:	Real-Time Stream Processing
Due:	As in Canvas
Weight:	5% of course grade
Format:	GitHub and README
Groups:	Individual submission only

Purpose

Build a Spark Structured Streaming pipeline using one of the four publicly available datasets below. Each scenario uses a different window type and maps to a distinct pattern from the lecture. Simulate streaming using Spark's reSadStream with a local file socket or watched directory.

What to Submit

- Your Spark Structured Streaming job code (github), a **README** file showing exactly how to run your project
- A short written explanation answering: *why you chose this window type, and where your pipeline requires state.*
- A screenshot of your alert output firing in the Spark console

Technical Requirements

- Use readStream with a file socket or watched directory
- Apply at least one window aggregation with withWatermark
- Define and trigger at least one alert condition as a filtered output stream

Choose One Scenario

Download the dataset for your chosen option before you begin. A sample of 10,000–50,000 rows is sufficient. You do not need to process the full dataset.

	Scenario	Dataset	Window	Task & alert condition
A	**Autonomous Fleet Telemetry** Detect sudden decelerations in connected vehicle streams that may indicate accidents.	AV GPS Dataset (GitHub)	Sliding 1 min / 10 s	Compute average speed per vehicle over the sliding window. Flag vehicles whose average drops >20 km/h between consecutive windows. **Alert:** *Speed drop >20 km/h in 10 s → anomaly with vehicle ID and timestamp.*

B	**Hospital Patient Monitoring** Detect sustained abnormal heart rates, not single spikes, across ICU patient streams.	IoMT Health Monitoring (Kaggle)	Tumbling 2 min	Compute average heart rate per patient per tumbling window. Flag patients exceeding 100 bpm in two consecutive windows. **Alert:** *Sustained elevated HR across 2 windows → clinical alert with patient ID.*
C	**E-Commerce Fraud Detection** Detect bot or fraud patterns in user clickstreams on an e-commerce platform.	eCommerce Behavior Data (Kaggle)	Session 30 min gap	Group events per user by session window. Flag users adding >5 items to cart within a session without a purchase event. **Alert:** *Cart-without-purchase in session → fraud flag with user ID and session start time.*
D	**Smart Power Grid** Detect residential zones unexpectedly exceeding industrial consumption levels.	Household Power Consumption (UCI)	Sliding 5 min / 1 min	Compute average consumption per zone. Enrich the stream by joining with a static CSV mapping meter IDs to zone type. Alert when a residential zone exceeds the industrial average. **Alert:** *Residential consumption > industrial average → grid anomaly with zone ID.*